

Random Vectors

A collection of n random variables

$X = (X_1, X_2, \dots, X_n)^T$ is referred to as a random vector.

Random matrix.

A collection of $m \times n$ random variables

$X = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & \vdots & & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$ is referred to as a random matrix

Expectation of a random matrix

$$E[X] = \begin{pmatrix} E[x_{11}] & E[x_{12}] & \dots & E[x_{1n}] \\ \vdots & \vdots & & \vdots \\ E[x_{m1}] & \dots & \dots & E[x_{mn}] \end{pmatrix}$$

Covariance matrix of a random vector

$$K_X = E \left[\underbrace{(X - E[X])}_{n \times 1} \underbrace{(X - E[X])^T}_{1 \times n} \right]$$

$$= E[\bar{X} \bar{X}^T]$$

$$= \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & & & \\ \vdots & & & \\ \text{Cov}(X_n, X_1) & - & - & \text{Cov}(X_n, X_n) \end{bmatrix}$$

$$\bar{X} = \begin{bmatrix} \bar{X}_1 \\ \vdots \\ \bar{X}_n \end{bmatrix}$$

where $\bar{X}_i = X_i - E[X_i]$

$$\bar{X} \bar{X}^T$$

$$= \begin{bmatrix} \bar{X}_1 \\ \vdots \\ \bar{X}_n \end{bmatrix} [\bar{X}_1 \dots \bar{X}_n] \\ = \begin{bmatrix} \bar{X}_1 \bar{X}_1 & \bar{X}_1 \bar{X}_2 & \dots & \bar{X}_1 \bar{X}_n \\ \vdots & \vdots & & \vdots \\ \bar{X}_n \bar{X}_1 & \dots & \dots & \bar{X}_n^2 \end{bmatrix}$$

Properties of the Covariance matrix

- ① K_x is symmetric.
- ② positive-semi-definite matrix ^(p.s.d)

A symmetric matrix $\underline{M}$ is said to be psd if $\forall \underline{a} \in \mathbb{R}^n$ $\underline{a}^T \underline{M} \underline{a} \geq 0$.

Spectral decomposition Theorem

Given any symmetric matrix M , it can be diagonalized using orthonormal matrices as

$$M = Q D Q^T \quad \text{where } Q \text{ is orthonormal matrix}$$

i.e., $Q Q^T = I$.

$$M Q = Q D$$

$$M \underline{q}_1 = \lambda_1 \underline{q}_1$$

$$M \underline{q}_i = \lambda_i \underline{q}_i$$

$$Q = [\underline{q}_1 \ \underline{q}_2 \ \dots \ \underline{q}_n]$$

$$D = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

Finding eigen values

$$M = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\det(M - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = 0 \quad \Rightarrow \quad (1-\lambda)^2 - 4 = 0$$
$$\lambda = 3, -1$$

$$Q D = [\underline{q}_1 \lambda_1 \ \underline{q}_2 \lambda_2 \ \dots]$$

$$M Q = [M \underline{q}_1 \ M \underline{q}_2 \ \dots]$$

Following defs are equivalent.

① M is s.t. $\underline{a}^T M \underline{a} \geq 0$.

② Eigen values of M are ≥ 0 .

③ $\exists$ a matrix A s.t. $M = AA^T$.

③ $\rightarrow$ ①

Suppose $\exists A$ s.t. $M = AA^T$.

$$\underline{a}^T AA^T \underline{a} = \underset{\substack{\downarrow \\ \text{let } y = A\underline{a}}}{y^T y} \geq 0.$$

② $\rightarrow$ ③

From spectral decomposition $M = QDQ^T$
where Q is orthonormal.
entries of D are ≥ 0 .

$$D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

$$D^{1/2} = \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix}$$

$$\begin{aligned} M &= QD^{1/2}D^{1/2}Q^T \\ &= \underbrace{QD^{1/2}Q^T}_A \underbrace{QD^{1/2}Q^T}_{A^T} \end{aligned}$$

① $\rightarrow$ ②

$$\text{Let } M \underline{q}_i = \lambda_i \underline{q}_i$$

From ①

$$\underline{a} = \underline{q}_i$$

$$\underline{q}_i^T M \underline{q}_i \geq 0.$$

$$= \underline{q}_i^T \lambda_i \underline{q}_i \geq 0.$$

$$= \lambda_i (\underline{q}_i^T \underline{q}_i) \geq 0$$

$$\Rightarrow \lambda_i \geq 0.$$

Jointly Gaussian RVs / Gaussian Vectors

$X = (x_1, \dots, x_n)^T$ is said to be zero mean Gaussian vector if $\exists$ a matrix A such that-

$$X = A W$$

$n \times m$ $m \times 1$

where $W = (w_1, \dots, w_m)^T$ is a collection of iid standard normal R.Vs.

$U = (u_1, \dots, u_n)^T$ is said to be Gaussian vector if $U = X + b$ where X is a zero mean Gaussian vector. where $b \in \mathbb{R}^n$.

$$\begin{aligned} f_W(\underline{w}) &= \prod_{i=1}^m f_{w_i}(w_i) \quad \text{as } w_i\text{'s are independent.} \\ &= \prod_{i=1}^m \left(\frac{e^{-w_i^2/2}}{\sqrt{2\pi}} \right) \end{aligned}$$

Let $Z = (z_1, \dots, z_n)^T$ be a zero mean Gaussian random vector and $Y = BZ$. Then Y is also zero-mean Gaussian.

Proof: Since Z is zero mean Gaussian

$\exists$ a matrix A st $Z = AW$
where W is standard normal R.V.

$$Y = \underbrace{B^T A^T}_C W$$

$$= CW$$

$\Rightarrow Y$ is a zero mean Gaussian vector.

Suppose Y is a Gaussian random vector then

$[a_1, a_2, \dots, a_n] Y$ is also a Gaussian random variable.

① Suppose Y_1 and Y_2 are Gaussian doesn't imply that they are jointly Gaussian.

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix}$$

② If two jointly Gaussian R.Vs are uncorrelated then they are independent.

③ if two Gaussian R.Vs are uncorrelated then it doesn't imply that are independent.